

LLM-Based Personalized Tutor for K-12 Short Answer Questions

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Abstract—Giving individualized and timely feedback to K-12 students on short-answer questions is a challenging job for teachers, particularly when working with large classes. This paper presents a comparative study of small language models for automatically generating age-appropriate educational feedback for students in Grades 3 to 8 in Mathematics and Science. We test five model configurations across 11 experimental setups: GPT-2 Small as a weak baseline, Flan-T5 Small under three zero-shot prompt strategies and one fine-tuned setting, Flan-T5 Base under three zero-shot prompt strategies, and mT5 Small under three zero-shot prompt strategies. A custom dataset of 100 samples was created with 13 misconception categories and expert-authored reference feedback. Models are assessed using ROUGE-1, ROUGE-2, ROUGE-L, and BLEU. Findings indicate that all zero-shot models fail to produce proper educational feedback. mT5 Small generates sentinel tokens making it completely unsuitable for this task. Fine-tuning Flan-T5 Small increases ROUGE-1 by 29.4% over its best zero-shot result but causes regression in ROUGE-2 and BLEU due to output collapse from insufficient training data. Flan-T5 Base with Prompt C achieves the best ROUGE-2 (0.0515) and BLEU (0.0183) in the zero-shot setting. A qualitative error analysis identifies four recurring failure modes: arithmetic hallucination, generic non-specific feedback, repetitive templating, and topically relevant but causally incorrect outputs. Our findings show that prompt design and model size both significantly affect feedback quality, and that fine-tuning on small datasets is not sufficient for generating reliable educational feedback.

Index Terms—personalized feedback, K-12 education, Flan-T5, mT5, GPT-2, prompt engineering, fine-tuning, ROUGE, BLEU, misconception detection, NLP in education

I. INTRODUCTION

Giving personalized feedback to each student after every short-answer question is incredibly difficult for teachers, particularly in large classes. Students who give wrong answers are usually made to wait a long time before being corrected, which slows down their learning process. This problem could be addressed by automated feedback systems powered by language models that can provide students with instant and useful explanations of their mistakes.

NLP has been growing significantly over the past few years. Large language models such as GPT-4 and LLaMA are very powerful but consume large amounts of computing resources to run. This makes them not viable for schools with limited budgets or in low-resource settings. Compared to larger models, small language models such as Flan-T5 Small (77 million parameters) are cheaper to run and can be fine-tuned

on small datasets, making them more practical for educational use.

This paper investigates whether small LLMs can generate meaningful feedback for K-12 students who give incorrect answers to short questions. We focus on two subjects, Math and Science, covering Grades 3 to 8. Our main contributions are:

- 1) A curated 100-sample K-12 short-answer dataset spanning Mathematics and Science for Grades 3 to 8, annotated with expert feedback and 13 misconception categories.
- 2) A systematic comparison of five model families under 11 experimental configurations including three prompt strategies for each zero-shot model.
- 3) Supervised fine-tuning of Flan-T5 Small on 80 training samples with evaluation of its impact on generation quality.
- 4) A detailed error analysis on 20 held-out samples cataloguing failure modes by misconception type, subject, and grade band.
- 5) Practical recommendations for future deployment of LLM-based feedback systems in K-12 settings.

II. PROBLEM STATEMENT

Given a K-12 short-answer question q , a student response s , the correct reference answer a^* , and metadata comprising the student's grade level g , subject σ , and diagnosed misconception type m , the task is to generate a feedback string f that:

- Correctly identifies the error in s relative to a^*
- Explains why the answer is incorrect in terms a student at grade level g can understand
- Guides the student toward a^* without simply stating it verbatim
- Maintains an encouraging and age-appropriate pedagogical tone

This is framed as a conditional text generation problem: learn a function F such that

$$\hat{f} = F(q, s, a^*, g, \sigma, m) \quad (1)$$

and $\hat{f}$ maximises similarity to ground-truth expert feedback f^* as measured by ROUGE and BLEU metrics, while also satisfying qualitative pedagogical criteria.

III. RESEARCH QUESTIONS

RQ1: Can a small LLM generate useful and accurate feedback for K-12 short-answer responses?

Hypothesis: Small LLMs can generate reasonably accurate feedback in some cases but will struggle with deeper conceptual understanding and complex student mistakes.

RQ2: How does prompt design affect the quality of feedback generated by the LLM?

Hypothesis: Structured prompts with role assignment and misconception metadata will significantly improve feedback quality over basic prompts.

RQ3: Does fine-tuning Flan-T5 Small on 80 training samples improve overall ROUGE scores relative to its zero-shot baseline?

Hypothesis: Fine-tuning will improve quantitative scores but may also lead to overfitting due to the small dataset size.

RQ4: What failure modes recur most frequently in generated feedback and do they cluster by subject, grade level, or misconception type?

Hypothesis: The model will fail more often on abstract or multi-step misconceptions than on concrete factual ones.

IV. RELATED WORK

A number of recent papers have considered the use of language models in education. The most relevant work is reviewed below.

Hicke et al. (2023) [1] evaluated GPT-4, fine-tuned GPT-2, and fine-tuned Flan-T5 for generating teacher responses in educational dialogues. Fine-tuned models trained with reinforcement learning performed better than zero-shot approaches. Their work is closely related to ours but focuses on dialogue rather than short-answer feedback.

Chen et al. (2024) [2] carried out a systematic review of prompt engineering for LLMs in K-12 STEM education covering 30 papers. They found that structured and few-shot prompting gives better results than simple zero-shot prompting, and that smaller fine-tuned models sometimes outperform larger models on specific educational tasks.

Stahl et al. (2025) [3] proposed SEFL, a framework for generating synthetic educational feedback using LLM agents. They showed that synthetic data can be effectively used for educational NLP tasks, which motivated our custom dataset creation approach.

Wang and Demszky (2025) [4] studied LLM-generated feedback in teacher education and found that structured theory-driven prompts consistently produce better feedback than simple prompts, directly supporting our hypothesis for RQ2.

Do Minh et al. (2025) [5] applied LLMs for educational management in a low-resource Vietnamese language setting. They showed that fine-tuning smaller models on domain-specific data gives better performance than using general large models in resource-limited environments, which motivates our choice of Flan-T5 Small as the primary fine-tuned model.

Kuklick et al. (2024) [6] integrated LLMs into Intelligent Tutoring Systems and evaluated motivational feedback generation using Llama 3. They found that LLM feedback quality depends heavily on model size and prompt design, showing the practical gap between large and small LLMs for educational tasks.

Qian (2025) [7] reviewed prompt engineering approaches in education and found that role-based structured prompts consistently outperform plain prompts, directly motivating our three-prompt experimental design.

Compared to all of this work, our study is the first to specifically investigate fine-tuning small open-source LLMs for K-12 short-answer feedback with explicit misconception labeling across multiple grade levels and two subjects. Most related work uses large proprietary models like GPT-4 or focuses on higher education.

V. METHODOLOGY

A. Model Selection

GPT-2 Small (124M parameters): A decoder-only autoregressive model [8] used as a weak baseline. GPT-2 is not instruction-tuned and cannot follow structured prompts, so it sets the performance floor against which encoder-decoder models are compared.

Flan-T5 Small (77M parameters): An encoder-decoder model instruction-tuned on a large collection of NLP tasks [9]. Tested in both zero-shot and fine-tuned settings. Its small size makes it the most practical candidate for low-resource educational deployment.

Flan-T5 Base (248M parameters): A larger version of Flan-T5 tested in zero-shot setting only. Used to compare the effect of model size versus fine-tuning on feedback quality.

mT5 Small (556M parameters): A multilingual variant of T5 pre-trained on 101 languages [10]. Although mT5 has more parameters than Flan-T5 Base, it was pre-trained with a masked span prediction objective making it fundamentally different from instruction-tuned models.

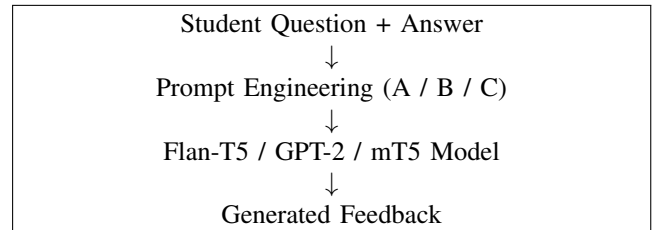


Fig. 1. Overview of the proposed feedback generation pipeline.

B. Prompt Engineering

Three prompt templates were designed to answer RQ2:

Prompt A (Basic): A minimal baseline with only the task instruction, question, student answer, and correct answer. No role or metadata.

Task: Write teacher feedback for a wrong student answer.
 Question: {q}
 Student Answer: {s}
 Correct Answer: {a*}
 Teacher Feedback:

Prompt B (Role-Based): Adds a teacher persona conditioned on the student grade level. Tests whether role assignment alone improves tone and clarity.

Task: You are a {grade} teacher. Write encouraging feedback for a student who answered wrongly.
 Question: {q}
 Student Answer: {s}
 Correct Answer: {a*}
 Teacher Feedback:

Prompt C (Structured with Misconception): The most information-rich prompt. Includes grade level, subject, and misconception type. Also used as the fine-tuning input format.

Task: Write teacher feedback for a wrong student answer.
 Grade: {g}
 Subject: {sigma}
 Misconception: {m}
 Question: {q}
 Student Answer: {s}
 Correct Answer: {a*}
 Teacher Feedback:

C. Fine-Tuning Setup

Supervised fine-tuning was performed on Flan-T5 Small using Prompt C as input and expert-authored feedback as the target sequence. A custom `FeedbackDataset` PyTorch class was implemented to tokenize input-target pairs. Label tokens corresponding to padding positions were set to -100 to exclude them from cross-entropy loss computation. The AdamW optimizer was paired with a linear learning rate schedule with warmup.

D. Generation Hyperparameters

All models used beam search decoding with `num_beams=4`, `no_repeat_ngram_size=3` to prevent repetition, `repetition_penalty=1.5`, and `length_penalty=2.0`. Maximum new tokens were capped at 150 for all models. GPT-2 required separate extraction of newly generated tokens by slicing from the input length since it generates continuations rather than independent sequences.

VI. DATASET

A. Construction

Since no existing dataset contained all required fields, we manually constructed a custom dataset of 100 samples. Each record contains: a unique item identifier, question text, erroneous student answer, correct reference answer, grade level (Grade 3 to Grade 8), subject (Mathematics or Science), diagnosed misconception type, and expert-authored expected feedback.

Expert feedback follows a consistent four-part pedagogical structure: (1) an encouraging opening such as “Good try!” or “Almost there!”, (2) identification of the specific mistake, (3) a simple explanation of the correct concept, and (4) a memory tip starting with “Remember:”. This structure was designed to match age-appropriate feedback practices for K-12 students. The dataset is publicly available at <https://www.kaggle.com/datasets/hafizabdullah66/k-12-student-feedback-dataset>.

B. Dataset Statistics

Table I summarises the distribution of records across subjects and grade levels.

TABLE I
DATASET DISTRIBUTION BY SUBJECT AND GRADE LEVEL

Category	Count
<i>Subject</i>	
Mathematics	52
Science	48
<i>Grade Level</i>	
Grade 3	11
Grade 4	28
Grade 5	25
Grade 6	23
Grade 7	12
Grade 8	1
Total	100

C. Misconception Taxonomy

The dataset covers 13 misconception categories providing fine-grained signal for targeted error analysis. Table II shows the full distribution.

D. Train/Test Split

For zero-shot evaluation all 100 samples were used since zero-shot models have not seen any training data. For supervised fine-tuning an 80/20 stratified split was applied giving 80 samples for training and 20 for held-out evaluation. The fine-tuned model is evaluated only on the 20 test samples to avoid data leakage.

TABLE II
MISCONCEPTION TYPE DISTRIBUTION

Misconception Type	Count
Fraction addition error	10
Basic algebra error	10
Multiplication and division confusion	9
Misunderstanding of photosynthesis	9
Area vs. perimeter confusion	8
Place value misunderstanding	8
Confusion between body organs	8
Habitat and adaptation confusion	8
Water cycle misunderstanding	7
Geometry angle confusion	7
Misunderstanding of states of matter	6
Food chain/ecosystem confusion	6
Gravity and force confusion	4
Total	100

VII. SYSTEM ARCHITECTURE

The system operates as a three-stage pipeline. In the *ingestion stage*, student responses are read from the dataset alongside contextual metadata including grade level, subject, and misconception type. In the *inference stage*, the selected model receives a formatted prompt and produces a feedback string via beam search decoding. In the *evaluation stage*, generated feedback is scored against reference feedback using automated metrics and a subset is subjected to qualitative error analysis.

Memory management between models is handled by explicit deletion using `del model` and `torch.cuda.empty_cache()` calls to avoid GPU out-of-memory errors when sequentially loading multiple model checkpoints on the Colab T4 GPU.

VIII. IMPLEMENTATION DETAILS

A. Software Stack

All experiments were implemented using Python 3 with PyTorch as the deep learning backend. HuggingFace Transformers was used for model loading and tokenization, the `evaluate` library for ROUGE metrics, and NLTK for BLEU computation using corpus-level BLEU with Method 1 smoothing. All experiments were conducted on Google Colab with an NVIDIA T4 GPU.

B. Hyperparameter Settings

Table III summarises the fine-tuning hyperparameter settings.

IX. EVALUATION METRICS

ROUGE-1 [11] measures unigram word-level overlap between generated and reference feedback. It captures how many individual words match.

ROUGE-2 measures bigram two-word phrase overlap. It checks whether the model produces correct phrases, not just isolated words.

TABLE III
FINE-TUNING HYPERPARAMETER SETTINGS

Hyperparameter	Value
Base Model	Flan-T5 Small
Epochs	10
Learning Rate	5e-5
Batch Size	2
Optimizer	AdamW
Scheduler	Linear with warmup
Max Input Length	512 tokens
Max Target Length	150 tokens
Padding Label	−100 (excluded from loss)
num_beams	4
no_repeat_ngram	3
repetition_penalty	1.5
length_penalty	2.0
max_new_tokens	150
Hardware	Google Colab NVIDIA T4 GPU

ROUGE-L measures the longest common subsequence between outputs and references. It captures structural similarity.

BLEU [12] measures precision of n-gram matches with a brevity penalty. It is included as a secondary metric. BLEU scores are generally low for short reference texts which is expected in our case.

ROUGE metrics are chosen as the primary evaluation because they are standard for text generation tasks and suitable for feedback evaluation where multiple valid responses may exist. Both ROUGE and BLEU have known limitations for short reference texts and should be interpreted alongside qualitative analysis.

X. RESULTS

A. Quantitative Results

Table IV presents ROUGE-1, ROUGE-2, ROUGE-L, and BLEU scores for all 11 model configurations.

A number of notable trends appear in Table IV. No zero-shot model scored higher than 0.23 on ROUGE-1. Fine-tuning Flan-T5 Small achieved the highest ROUGE-1 of 0.2400, which is a 29.4% improvement over its best zero-shot result. mT5 Small produced BLEU scores of exactly 0.0000 for all prompts due to sentinel token outputs. Flan-T5 Base consistently outperformed Flan-T5 Small in zero-shot, demonstrating the advantage of larger model size. Flan-T5 Base Prompt C achieved the best ROUGE-2 (0.0515) and the highest BLEU (0.0183) among all configurations.

B. GPT-2 Small — Weak Baseline

GPT-2 achieves ROUGE-1 of 0.1657 and BLEU of 0.0058, confirming its role as a weak baseline. Qualitative inspection shows that GPT-2 generates plausible-sounding continuations that quickly move away from the educational task. Outputs frequently invent fake teacher-student conversations, list tangentially related facts, or produce random numbered lists of objects. Since GPT-2 is decoder-only without instruction

TABLE IV
ROUGE AND BLEU SCORES FOR ALL MODEL CONFIGURATIONS

Model / Configuration	ROUGE-1	ROUGE-2	ROUGE-L	BLEU
GPT-2 Small (Baseline)	0.1657	0.0256	0.1076	0.0058
Flan-T5 Small ZS Prompt A	0.1724	0.0426	0.1142	0.0165
Flan-T5 Small ZS Prompt B	0.1538	0.0366	0.1017	0.0076
Flan-T5 Small ZS Prompt C	0.1855	0.0504	0.1285	0.0113
Flan-T5 Small Fine-Tuned*	0.2400	0.0325	0.1251	0.0047
Flan-T5 Base ZS Prompt A	0.2162	0.0491	0.1361	0.0181
Flan-T5 Base ZS Prompt B	0.2246	0.0461	0.1383	0.0189
Flan-T5 Base ZS Prompt C	0.2292	0.0515	0.1374	0.0183
mT5 Small ZS Prompt A	0.0726	0.0287	0.0652	0.0000
mT5 Small ZS Prompt B	0.0799	0.0266	0.0704	0.0000
mT5 Small ZS Prompt C	0.0639	0.0173	0.0565	0.0000

*Fine-tuned model evaluated on 20 test samples. All zero-shot models evaluated on 100 samples.

tuning, it cannot understand or follow the teacher-feedback task.

C. Flan-T5 Small: Zero-Shot vs. Fine-Tuned

Among Flan-T5 Small zero-shot configurations, Prompt C achieves the highest scores, confirming that richer metadata such as grade level, subject, and misconception type guides the model toward more relevant output. Prompt B performs worst among the three, suggesting that role-assignment alone without explicit misconception context does not add value for this model size.

After fine-tuning on 80 samples, Flan-T5 Small attains the highest ROUGE-1 (0.2400) among all configurations. However, ROUGE-2 drops to 0.0325 and BLEU falls to 0.0047, both below its zero-shot Prompt C baseline. This indicates that fine-tuning improved unigram overlap but the model began overfitting surface-level vocabulary from training labels while losing the ability to generate fluent multi-word sequences.

D. Flan-T5 Base — Best Zero-Shot Model

Flan-T5 Base with Prompt C achieves the best ROUGE-2 (0.0515) and a competitive ROUGE-1 (0.2292). Its BLEU score of 0.0183 is also the highest among all configurations. The larger parameter count of 248M versus 77M provides stronger instruction-following capability in zero-shot settings, allowing Prompt C’s structured misconception metadata to be effectively utilized without any fine-tuning.

E. mT5 Small — Sentinel Token Failure

mT5 Small performs dramatically worse than all other models, with ROUGE-1 values between 0.0639 and 0.0799 and BLEU of exactly 0.0000 across all three prompts. Qualitative inspection confirms that mT5 exclusively produces sentinel tokens such as <extra_id_0> followed by brief fragments of the input. This is a known failure mode when mT5 is prompted in a text-generation rather than span-infilling paradigm, making it completely unsuitable for this task without substantial task-specific adaptation.

F. Training Loss

Table V shows the per-epoch training loss for Flan-T5 Small fine-tuning.

TABLE V
FLAN-T5 SMALL FINE-TUNING TRAINING LOSS PER EPOCH

Epoch	Training Loss
1	9.5083
2	9.0843
3	8.8687
4	8.7357
5	8.6351
6	8.5503
7	8.4869
8	8.4458
9	8.4224
10	8.4062

Training loss decreases steadily from 9.5083 to 8.4062 over 10 epochs. While the loss decreases consistently, the absolute values remain high, indicating the model did not fully converge, mainly because of the very small training set of 80 samples.

XI. ANALYSIS

A. Error Category Distribution

Table VI shows the distribution of output length-based error categories across all model configurations.

Flan-T5 Small in zero-shot produced incomplete outputs in 30 to 36 percent of cases. Fine-tuning completely resolved this issue, with all 20 outputs being longer than 10 words. Flan-T5 Base with Prompt C generated outputs longer than 10 words in 90 percent of cases. mT5 Small produced sentinel tokens in 100 percent of all outputs regardless of prompt.

B. Qualitative Failure Mode Analysis

Failure Mode 1 — Arithmetic Hallucination: The most damaging failure mode. The model generates fabricated numerical expressions unrelated to the problem. For the algebra

TABLE VI
ERROR CATEGORY DISTRIBUTION ACROSS MODELS

Model	<5 words	5–10 words	>10 words	Sentinel	N
GPT-2 Small	0 (0%)	0 (0%)	100 (100%)	0 (0%)	100
Flan-T5 Small ZS Prompt A	30 (30%)	17 (17%)	53 (53%)	0 (0%)	100
Flan-T5 Small ZS Prompt B	36 (36%)	18 (18%)	46 (46%)	0 (0%)	100
Flan-T5 Small ZS Prompt C	21 (21%)	28 (28%)	51 (51%)	0 (0%)	100
Flan-T5 Small Fine-Tuned	0 (0%)	0 (0%)	20 (100%)	0 (0%)	20
Flan-T5 Base ZS Prompt A	12 (12%)	8 (8%)	80 (80%)	0 (0%)	100
Flan-T5 Base ZS Prompt B	4 (4%)	19 (19%)	77 (77%)	0 (0%)	100
Flan-T5 Base ZS Prompt C	1 (1%)	9 (9%)	90 (90%)	0 (0%)	100
mT5 Small (All Prompts)	0 (0%)	0 (0%)	0 (0%)	100 (100%)	100

question “Solve: $x + 9 = 20$ ” (correct: $x = 11$), the fine-tuned model generated: “*It is a total of 1 and 2. The number in the first place will be 3: $5 + 4 = 9!$* ” The numbers have no mathematical relation to the problem. This is particularly harmful because it risks reinforcing the student’s misconception rather than correcting it.

Failure Mode 2 — Generic Non-Specific Feedback: Outputs are syntactically valid but semantically empty. For the question “What is hibernation?” the model generated: “*It is a state of the air. You are not in it and you have to do so!*” This completely fails to address the student’s confusion between hibernation and migration.

Failure Mode 3 — Repetitive Templated Phrasing: The fine-tuned model converges on a small set of sentence templates such as “*It is a [noun]. You are [not/in] it...*” across multiple outputs. This indicates the model memorised high-frequency surface patterns from 80 training samples rather than learning the underlying feedback task.

Failure Mode 4 — Correct Topic, Wrong Content: In some Science questions the model correctly identifies the topic domain but inverts or distorts the causal relationship. For “Why do cacti have thick stems?” the model retrieved desert-related vocabulary but produced: “*They will be more than other plants, so they may have as long or less when there’s just one plant at an area on your body.*” Topically relevant but causally meaningless.

C. Analysis by Subject and Grade Level

Table VII shows ROUGE scores broken down by subject and grade level for the fine-tuned Flan-T5 Small model.

TABLE VII
ROUGE SCORES BY SUBJECT AND GRADE LEVEL
(FINE-TUNED FLAN-T5 SMALL, 20 TEST SAMPLES)

Category	R-1	R-2	R-L
Mathematics	0.2625	0.0369	0.1443
Science	0.2240	0.0297	0.1133
Grade 3–5 (Lower)	0.2481	0.0346	0.1273
Grade 6–8 (Upper)	0.2045	0.0246	0.1196

The fine-tuned model performed better on Math questions (ROUGE-1: 0.2625) than Science (ROUGE-1: 0.2240). Math

feedback uses more structured and repetitive language such as step-by-step calculation instructions which the model may have learned more easily. For grade level, the model performed better on lower grade samples (ROUGE-1: 0.2481) than upper grade samples (ROUGE-1: 0.2045) because lower grade feedback uses simpler vocabulary.

D. Ablation Study: Zero-Shot vs. Fine-Tuned

Table VIII presents the ablation study comparing zero-shot versus fine-tuned performance of Flan-T5 Small.

TABLE VIII
ABLATION STUDY — EFFECT OF FINE-TUNING (FLAN-T5 SMALL)

Configuration	R-1	R-2	R-L	BLEU
ZS Prompt A	0.1724	0.0426	0.1142	0.0165
ZS Prompt B	0.1538	0.0366	0.1017	0.0076
ZS Prompt C (Best)	0.1855	0.0504	0.1285	0.0113
Fine-Tuned	0.2400	0.0325	0.1251	0.0047
Change vs Best ZS	+29.4%	−35.5%	−2.6%	−58.4%

Fine-tuning improved ROUGE-1 by 29.4% but caused significant regression in ROUGE-2 (−35.5%) and BLEU (−58.4%). This is a classic sign of surface-level overfitting on a small dataset. The fine-tuned model learned which individual words to use but lost the ability to generate correct phrases or sentence structures.

E. Misconception-Level Performance

Error analysis reveals that misconceptions involving concrete factual corrections such as gravity and force confusion (ROUGE-L: 0.1839) and multiplication errors (ROUGE-L: 0.1443) achieved higher scores than abstract multi-step misconceptions such as basic algebra error (ROUGE-L: 0.1349) and food chain confusion (ROUGE-L: 0.0770). This confirms our hypothesis for RQ4: models perform better on concrete factual corrections than on abstract multi-step reasoning errors.

XII. CHALLENGES AND LIMITATIONS

Small Training Corpus: The most important limitation is that only 80 fine-tuning samples were used. State-of-the-art

fine-tuning for text generation typically requires thousands of examples. The current dataset is not sufficient for stable convergence, leading to output collapse and hallucination.

Metric-Pedagogy Gap: ROUGE and BLEU measure surface-level lexical overlap with a single reference feedback string. Pedagogically valid feedback can be expressed in many different ways. A model generating a different but equally correct explanation will be penalised by these metrics. Human evaluation was planned but not completed, leaving the gap between automated metrics and genuine pedagogical quality unquantified.

Single Reference Per Sample: Each dataset item has exactly one expert-authored reference feedback string. ROUGE scores computed against a single reference underestimate model quality when the model generates an alternative but equally valid explanation [11].

mT5 Prompt Incompatibility: The mT5 model’s span-corruption pre-training objective makes it fundamentally unsuited to free-form text generation. Its inclusion effectively served as a negative control rather than a genuine competitor.

No Age-Appropriateness Verification: While Prompt C conditions on grade level, no systematic mechanism verifies that generated vocabulary and sentence complexity are actually appropriate for the target grade.

XIII. CONCLUSION

This paper presented a comparative study of LLM-based feedback generation for K-12 short-answer questions. We evaluated 11 configurations of four model families across a 100-sample custom dataset with 13 annotated misconception categories.

Answering our research questions: For **RQ1**, Flan-T5 Base with Prompt C achieves the best overall balance of ROUGE-2 (0.0515) and BLEU (0.0183) making it the most suitable zero-shot configuration. However, all models fell far short of human-level feedback quality. For **RQ2**, Prompt C consistently outperforms Prompts A and B confirming that structured misconception-aware prompts improve feedback quality. For **RQ3**, fine-tuning Flan-T5 Small on 80 samples improves ROUGE-1 by 29.4% but causes severe regression in ROUGE-2 and BLEU due to surface-level overfitting on insufficient training data. For **RQ4**, failure modes cluster by misconception type with concrete factual errors yielding better scores than abstract multi-step reasoning errors.

XIV. FUTURE WORK

Larger Models: Scaling to Flan-T5 Large (780M) or using parameter-efficient fine-tuning methods like LoRA could make larger models accessible in resource-constrained settings.

Expanded Dataset: A dataset of 500 to 1000 samples with multiple reference feedback strings per item would enable more reliable evaluation and stable fine-tuning.

Human Evaluation: A structured rubric assessing factual accuracy, age-appropriateness, encouraging tone, and misconception specificity should be the primary evaluation signal.

Grade-Adaptive Language Control: Incorporating readability constraints such as Flesch-Kincaid grade level targets during generation would ensure feedback vocabulary is appropriate for the target grade band.

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